Ayush Jain

Research Scientist, Applied Reinforcement Learning
Meta, Monetization Org

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Education

University of Southern California Dec 2024
PhD and MS in Computer Science (Advisors: Joseph J. Lim and Erdem Bıyık) GPA: 4.0/4.0

Indian Institute of Technology Delhi Jun 2016
B.Tech in Electrical Engineering GPA: 8.99/10

Research Interests and Industrial Impact

PhD Work

Reinforcement-learning algorithms for agents that reason over large, structured, unseen, or otherwise complex action spaces. **PhD thesis:** Decision Making in Complex Action Spaces.

Applied RL at Meta

- Offline RL for long term value optimization in ad recommender-system.
- Contextual bandits for ad auctions and product exploration.
- RL post-training and evaluation for multi-turn LLM conversation assistants.

Keywords: offline reinforcement learning, recommender systems, robotics, LLM post-training, LLM multi-turn evaluation, instruction-following digital agents.

Professional Experience

Meta, Sunnyvale, CA Feb 2025 – Present
Research Scientist, Applied Reinforcement Learning Team, Monetization Org

Meta Reality Labs Research, Redmond, WA May 2024 – Dec 2024
Research Intern with Nitin Kamra

- RL-based digital agents for instruction-following in Android device environments.

Microsoft Research, Montreal, Canada May 2023 – Jan 2024
Research Intern with Eric Yuan and Marc-Alexandre Côté

- Discrete prompt optimization of LLMs in instruction-following Minecraft agents.

Naver AI Research, Seoul, South Korea (remote) 2021, 2022
Research Intern with Kyung-Min Kim and Joseph J. Lim

- RL for recommender systems with large, varying, and slate action spaces.

Samsung Research, Seoul, South Korea Sep 2016 – Jun 2018
Engineer, Data Analytics Lab with James Geraci and Yunsu Lee

- Time series modeling for (i) market share prediction and (ii) anomaly detection.

Samsung Electronics, Suwon, South Korea May 2015 – Jul 2015
Software Engineer Intern with Sungmok Seo

Publications

Peer-Reviewed Conference Papers

[C8] ICLR 2026. X. Li, **A. Jain**, Z. Yang, Y. Korkmaz, E. Bıyık. "When a Robot is More Capable than a Human: Learning from Constrained Demonstrators". *International Conference on Learning Representations.*

[C7] ICML 2026. A. Samanta, A. Magesh, **A. Jain**, K. Asadi, Y. Yu, D. Jiang, B. Vidolov, K. Hassani, P. Sajda, J. Bhandari, Y. Efroni. "Structure Enables Effective Self-Localization of Errors in LLMs". *International Conference on Machine Learning.*

[C6] EACL 2026. R. Wu, A. Samanta, **A. Jain**, S. Fujimoto, J. Kwon, B. Kretzu, Y. Yu, K. Hassani, B. Vidolov, Y. Efroni. "Imbalanced Gradients in RL Post-Training of Multi-Task LLMs". *19th Conference of the European Chapter of the Association for Computational Linguistics.*

[C5] NeurIPS 2025. Y. Korkmaz*, U. Bhuwania*, **A. Jain**†, E. Bıyık†. "Actor-Free Continuous Control via Structurally Maximizable Q-Functions". *Neural Information Processing Systems.*

[C4] RLC 2025. **A. Jain**, N. Kosaka, X. Li, K. Kim, E. Bıyık, J. Lim. "Mitigating Suboptimality of Deterministic Policy Gradients in Complex Q-functions". *Reinforcement Learning Conference.* **Outstanding Paper Award on Empirical Reinforcement Learning Research**.

[C3] ICLR 2025. G. Zhang*, **A. Jain***, I. Hwang, S. Sun, J. Lim. "QMP: Q-switch Mixture of Policies for Multi-Task Behavior Sharing". *International Conference on Learning Representations.*

[C2] ICLR 2022. **A. Jain***, N. Kosaka*, K. Kim, J. Lim. "Know Your Action Set: Learning Action Relations for Reinforcement Learning". *International Conference on Learning Representations.*

[C1] ICML 2020. **A. Jain***, A. Szot*, J. Lim. "Generalization to New Actions in Reinforcement Learning". *International Conference on Machine Learning.*

Preprints, Workshops, and Earlier Papers

[W3] RSS 2026 Workshop. R. Lindeborg, R. J. Peng, **A. Jain**, J. Zhang, A. Anwar, J. Thomason. "Anticipating Unintended Robot Behaviors with Side Effect Critics". *Rethinking What It Means to be "Safe" for Generalist Robots.* **Spotlight presentation**.

[W2] COLING 2018 Workshop. **A. Jain**, V. Singh, S. Ranjan, R. Rajkumar, S. Agarwal. "Uniform Information Density Effects on Syntactic Choice in Hindi". *Workshop on Linguistic Complexity and NLP.*

[W1] DGfS 2017. **A. Jain**, V. Singh, S. Agarwal, R. Rajkumar. "Uniform Information Density models for language production". *39th Annual Conference of the German Linguistic Society.*

[P3] arXiv. A. Samanta, A. Magesh, T. Lancewicki, **A. Jain**, Y. Yu, P. Sajda, K. Hassani, A. Modi, D. Jiang, Y. Efroni. "BayesBench: Evaluating LLM Belief Trajectories Under Multi-Turn Evidence Accumulation".

[P2] arXiv. A. Samanta, A. Magesh, **A. Jain**, Y. Yu, D. Jiang, K. Asadi, K. Hassani, P. Sajda, J. Bhandari, Y. Efroni. "Credit Assignment with Resets in Language Model Reasoning".

[P1] arXiv. A. Samanta, A. Magesh, R. Wu, **A. Jain**, Y. Yu, D. Jiang, B. Vidolov, P. Sajda, Y. Efroni, K. Hassani. "Self-Improvement of Language Models by Post-Training on Multi-Agent Debate".

Patent

[T1] J. Lee, M. Kim, **A. Jain**, T. Hwang, J. Kim, H. Cho. "Method and Apparatus for Managing Operation Data of Appliance for Failure Prediction". U.S. Patent No. 11,182,235. Nov 23, 2021.

Research Mentoring

Norio Kosaka (Naver AI → Ph.D. student at NII Japan)	ICLR 2022, RLC 2025
Xinhu Li (USC → Ph.D. student at NUS)	RLC 2025, ICLR 2026
Urvi Bhuwania (USC)	NeurIPS 2025
Zhaoyi Zhou (CMU, Meta research intern)	Current

Teaching and Service

Teaching Assistant, University of Southern California

Deep Learning and its Applications, CSCI-566.

- Fall 2019; Spring 2019; Fall 2020; Spring 2023; Spring 2024; Fall 2024.

Reviewer Service

- ICLR: 2023, 2024, 2025, 2026
- NeurIPS: 2023, 2024, 2025, 2026
- NeurIPS Workshop Proposals: 2026
- ICML: 2025, 2026
- CoRL: 2021, 2022, 2023, 2024
- RLC: 2025, 2026
- AAAI: 2026

Selected Honors

- Outstanding Paper Award on Empirical Reinforcement Learning Research (RLC 2025).
- Silver Reviewer Award, International Conference on Machine Learning (ICML 2026).
- Travel Grant, Reinforcement Learning Conference (RLC 2025).
- Director's Award for ranking in the top 7% at IIT Delhi (2013–2014).
- All India Rank 198 in IITJEE and All India Rank 91 in AIEEE (2012).
- KVPY Fellowship Award and National Talent Search Examination Scholarship Award.